

Homework 2

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Report Questions

1 Maximum Vertical Displacement

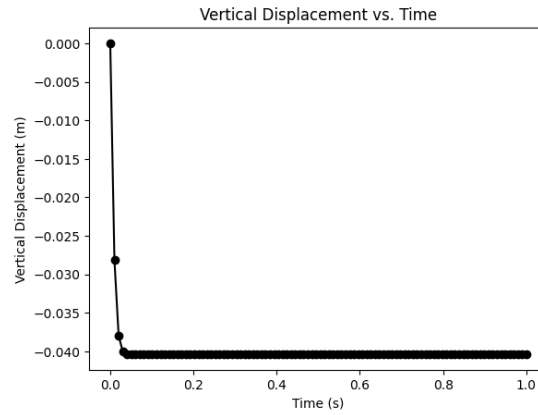


Figure 1: Maximum vertical displacement ($y_{\max}$) over time.

The vertical displacement $y_{\max}$ reaches a steady value approximately 0.05 seconds into the simulation. To compare this with the theoretical result from Euler-Bernoulli beam theory, the maximum deflection is given by

$$y_{\max, \text{theory}} = \frac{P c (l^2 - c^2)^{3/2}}{9\sqrt{3} E I l}, \quad c = \min(d, l - d),$$

where I is the second moment of area of the circular-tube cross-section:

$$I = \frac{\pi}{4} (R^4 - r^4).$$

Using the given parameters:

$$l = 1.0 \text{ m}, \quad R = 0.013 \text{ m}, \quad r = 0.011 \text{ m}, \quad E = 70 \times 10^9 \text{ Pa}, \quad d = 0.75 \text{ m},$$

we find $c = \min(0.75, 0.25) = 0.25$ m and

$$I = \frac{\pi}{4} (0.013^4 - 0.011^4) = 1.093 \times 10^{-8} \text{ m}^4.$$

For a load of $P = 2000$ N:

$$y_{\max, \text{theory}}(2000 \text{ N}) = 38.045 \text{ mm}.$$

The theoretical value (38.04 mm) is reasonably close to the simulation result (40 mm), demonstrating good agreement and validating the simulation's accuracy for small deformations.

2 Comparison Between Simulation and Beam Theory

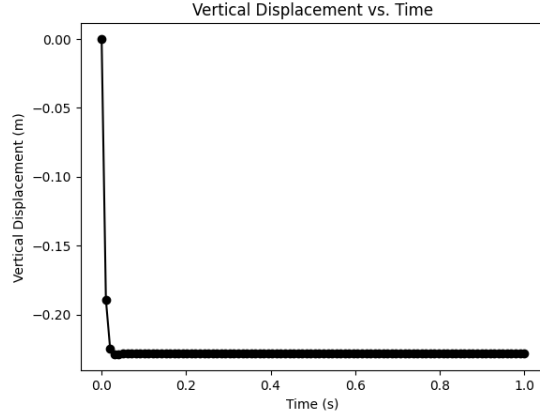


Figure 2: Maximum vertical displacement over time for $P = 20,000$ N, reaching 0.228 m.

A key advantage of numerical simulation over analytical beam theory is its ability to accurately capture large deformations without numerical instability or unrealistic distortion.

Using the Euler-Bernoulli beam theory for $P = 20,000$ N:

$$y_{\max, \text{theory}}(20,000 \text{ N}) = 0.38045 \text{ m}.$$

This differs significantly from the simulated deflection (0.228 m), highlighting the limitations of linear beam theory at high loads, where geometric nonlinearity becomes important.

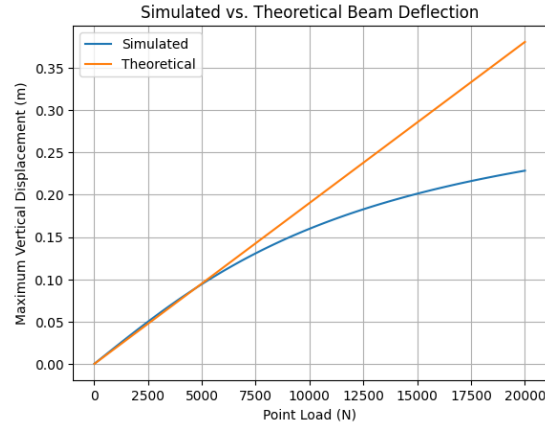


Figure 3: Comparison between simulated and theoretical beam deflection as a function of applied load.

As shown above, the Euler-Bernoulli theory provides accurate predictions up to approximately 6000 N. Beyond this range, it increasingly overestimates deflection, while the simulation continues to produce physically consistent results.

References

- [1] M. K. Jawed, “Lecture 5–Modules 10–12,” Dept. of Mechanical and Aerospace Engineering, Univ. of California, Los Angeles, 2025. [Class hand-out].
- [2] M. K. Jawed, “Lecture 6–Modules 13–20,” Dept. of Mechanical and Aerospace Engineering, Univ. of California, Los Angeles, 2025. [Class resource].
- [3] M. K. Jawed, “Simulation of a Beam with Fixed DOFs,” Dept. of Mechanical and Aerospace Engineering, Univ. of California, Los Angeles, 2025. [Class resource].